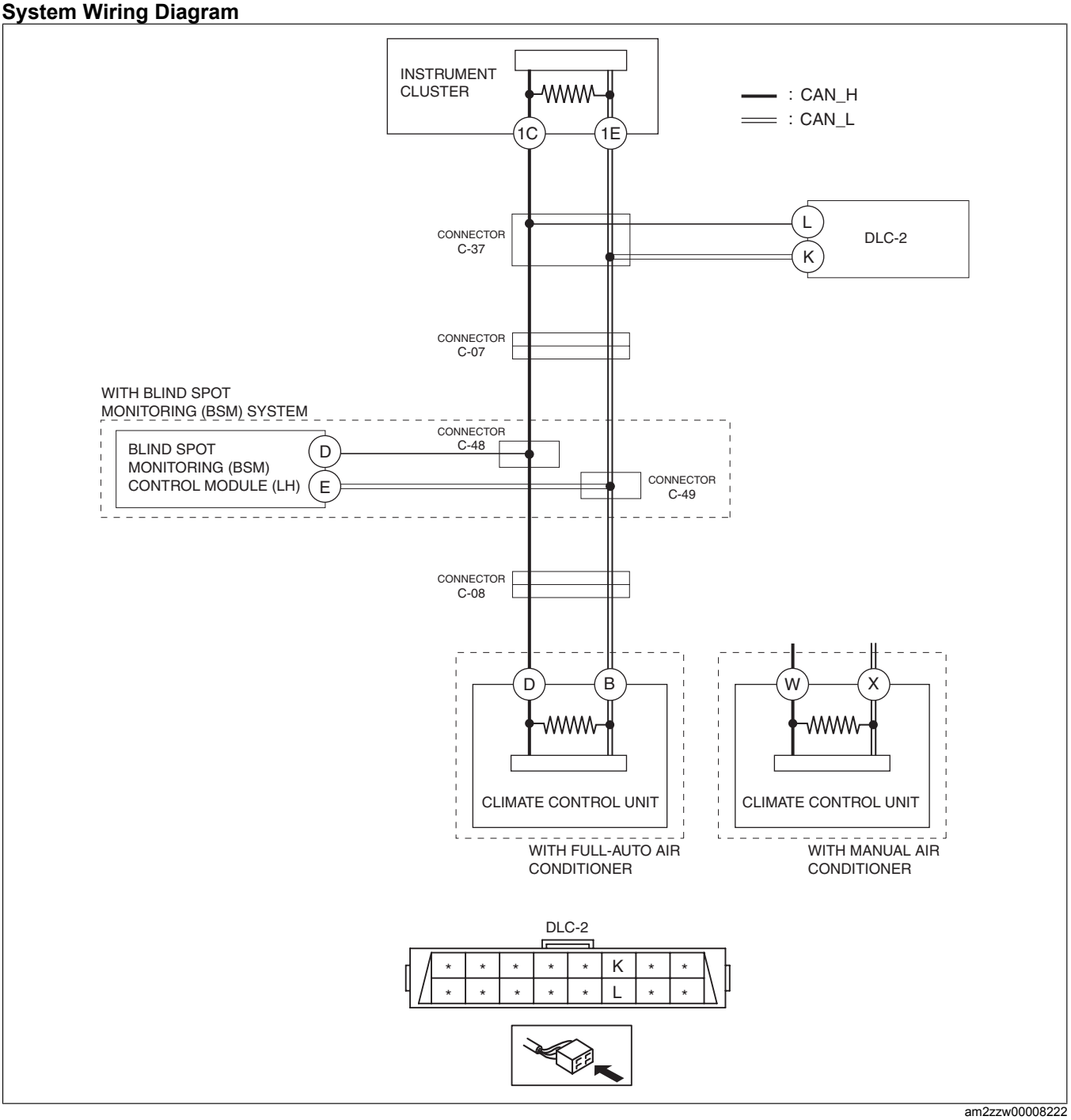
DETERMINING SHORT BETWEEN CIRCUITS LOCATION (MS-CAN) [SKYACTIV-G 1.3, SKYACTIV-G 1.5 (R.H.D.)]

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Caution

- **Perform the following malfunction diagnosis only when it is diagnosed with a short between circuits by CONTROLLER AREA NETWORK (CAN) MALFUNCTION DIAGNOSIS FLOW. (See CONTROLLER AREA NETWORK (CAN) MALFUNCTION DIAGNOSIS FLOW [SKYACTIV-G 1.3, SKYACTIV-G 1.5 (R.H.D.)].)**

System Wiring Diagram



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Determination Procedure

Caution

- **When disconnecting the connector, verify that there is no looseness, damage, deformation, corrosion, or poor connection of the connector terminals.**

- **When inspecting the DLC-2, touch it with a paper clip or similar thin pin without directly inserting a tester into the terminals.**

Step	Inspection		Action
1	**INSPECT BETWEEN INSTRUMENT CLUSTER AND CONNECTOR C-37 FOR SHORT BETWEEN CIRCUITS** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect connector C-37. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off). • Measure the voltage at instrument cluster terminals 1C and 1E. • Is the voltage at instrument cluster terminals 1C and 1E the same?	Yes	Go to the next step.
		No	Go to Step 3.
2	**INSPECT INSTRUMENT CLUSTER FOR SHORT BETWEEN CIRCUITS** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the instrument cluster connector. • Inspect for continuity between instrument cluster terminals C and E (wiring harness side). • Is there continuity?	Yes	Repair or replace the wiring harness between the instrument cluster and connector C-37 because the wiring harness is shorted between circuits.
		No	Replace the instrument cluster because there is a short between circuits in the instrument cluster. (See INSTRUMENT CLUSTER REMOVAL/ INSTALLATION.)
3	**INSPECT BETWEEN CONNECTOR C-37 AND DLC-2 FOR SHORT BETWEEN CIRCUITS** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Inspect for continuity between DLC-2 terminals L and K. • Is there continuity?	Yes	Repair or replace the wiring harness between connector C-37 and DLC-2 because the wiring harness is shorted between circuits.
		No	Go to the next step.
4	**INSPECT BETWEEN CONNECTOR C-07 AND CONNECTOR C-37 FOR SHORT BETWEEN CIRCUITS** • Disconnect connector C-07. • Connect connector C-37. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off). • Measure the voltage at DLC-2 terminals L and K. • Is the voltage at DLC-2 terminals L and K the same?	Yes	Repair or replace the wiring harness between connector C-07 and connector C-36 because the wiring harness is shorted between circuits.
		No	Go to the next step.

Step	Inspection		Action
5	**INSPECT BETWEEN CONNECTORS C-48 AND C-49 AND CONNECTOR C-07 FOR SHORT BETWEEN CIRCUITS** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect connectors C-48 and C-49. • Connect connector C-07. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off). • Measure the voltage at DLC-2 terminals L and K. • Is the voltage at DLC-2 terminals L and K the same?	Yes	Repair or replace the wiring harness between connectors C-48 and C-49 and connector C-07 because the wiring harness is shorted between circuits.
		No	Go to the next step.
6	**INSPECT BETWEEN BLIND SPOT MONITORING (BSM) CONTROL MODULE (LH) AND CONNECTORS C-48 AND C-49 FOR SHORT BETWEEN CIRCUITS** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Inspect the continuity between blind spot monitoring (BSM) control module (LH) terminals D and E. • Is there continuity?	Yes	Go to the next step.
		No	Go to Step 8.
7	**INSPECT BLIND SPOT MONITORING (BSM) CONTROL MODULE (LH) FOR SHORT BETWEEN CIRCUITS** • Disconnect the blind spot monitoring (BSM) control module (LH) connector. • Inspect for continuity between blind spot monitoring (BSM) control module (LH) terminals D and E (wiring harness side). • Is there continuity?	Yes	Repair or replace the wiring harness between the blind spot monitoring (BSM) control module (LH) and connectors C-48 and C-49 because the wiring harness is shorted between circuits.
		No	Replace the blind spot monitoring (BSM) control module (LH) because there is a short between circuits in the blind spot monitoring (BSM) control module (LH). (See BLIND SPOT MONITORING (BSM) CONTROL MODULE REMOVAL/INSTALLATION.)
8	**INSPECT BETWEEN CONNECTOR C-08 AND CONNECTORS C-48 AND C-49 FOR SHORT BETWEEN CIRCUITS** • Disconnect connector C-08. • Connect connectors C-48 and C-49. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off). • Measure the voltage at DLC-2 terminals L and K. • Is the voltage at DLC-2 terminals L and K the same?	Yes	Repair or replace the wiring harness between connector C-08 and connectors C-48 and C-49 because the wiring harness is shorted between circuits.
		No	Go to the next step.

Step	Inspection		Action
9	**INSPECT CLIMATE CONTROL UNIT FOR SHORT BETWEEN CIRCUITS** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the climate control unit connector. • Measure the voltage at climate control unit terminals D and B. (with full-auto air conditioner) • Measure the voltage at climate control unit terminals W and X. (with manual air conditioner) • Is the voltage at climate control unit terminals D and B the same? (with full-auto air conditioner) • Is the voltage at climate control unit terminals W and X the same? (with manual air conditioner)	Yes	Repair or replace the wiring harness between the climate control unit and connector C-08 because the wiring harness is shorted between circuits.
		No	Replace the climate control unit because there is a short between circuits inside the climate control unit. (See CLIMATE CONTROL UNIT REMOVAL/ INSTALLATION [FULL-AUTO AIR CONDITIONER].) (See CLIMATE CONTROL UNIT REMOVAL/ INSTALLATION [MANUAL AIR CONDITIONER].)